

APPLIED MATH FOR COMPUTER SCIENCE

CPCS 212

Lab 5: **Roots of Equations: Bisection Method**

A Sample Problem

Suppose we want to know if there is a solution to

$$\cos x = x,$$

Exercise 1:

How to use Matlab to plot the functions $y = \cos x$ and $y = x$ together in the same graph from $-\pi$ to π .

Open the diary file
Lab5_yourName.txt

1-(a)

Define a Matlab (vector) variable *xplot* to take on 100 evenly-spaced values between $-\pi$ and π . You can use *linspace* to do this.

```
>> xplot=linspace(-pi,pi,100);
```

1-(b)

Define the (vector) variable $y1plot$ by $y1plot = \cos(xplot)$.

This is for the curve $y = \cos x$.

```
>> y1plot=cos(xplot);
```

1-(c)

Define the (vector) variable $y2plot$ by $y2plot = xplot$.

This is for the line $y = x$.

```
>> y2plot=xplot;
```

1-(d)

Plot both *y1plot* and *y2plot* by using:

```
>> plot(xplot,y1plot)
```

```
>> hold on
```

```
>> plot(xplot,y2plot)
```

```
>> hold off
```


1-(e)

What would happen if the two lines *hold on* and *hold off* were omitted?

```
>> % alternative solution
```

```
>> plot(xplot,y1plot,xplot,y2plot)
```

1-(f)

Save the plot in the form of a jpg file.

```
>> print -djpeg e1lab5.jpg
```

e1Lab5.m

```
xplot=linspace(-pi,pi,100);
```

```
y1plot=cos(xplot);
```

```
y2plot=xplot;
```

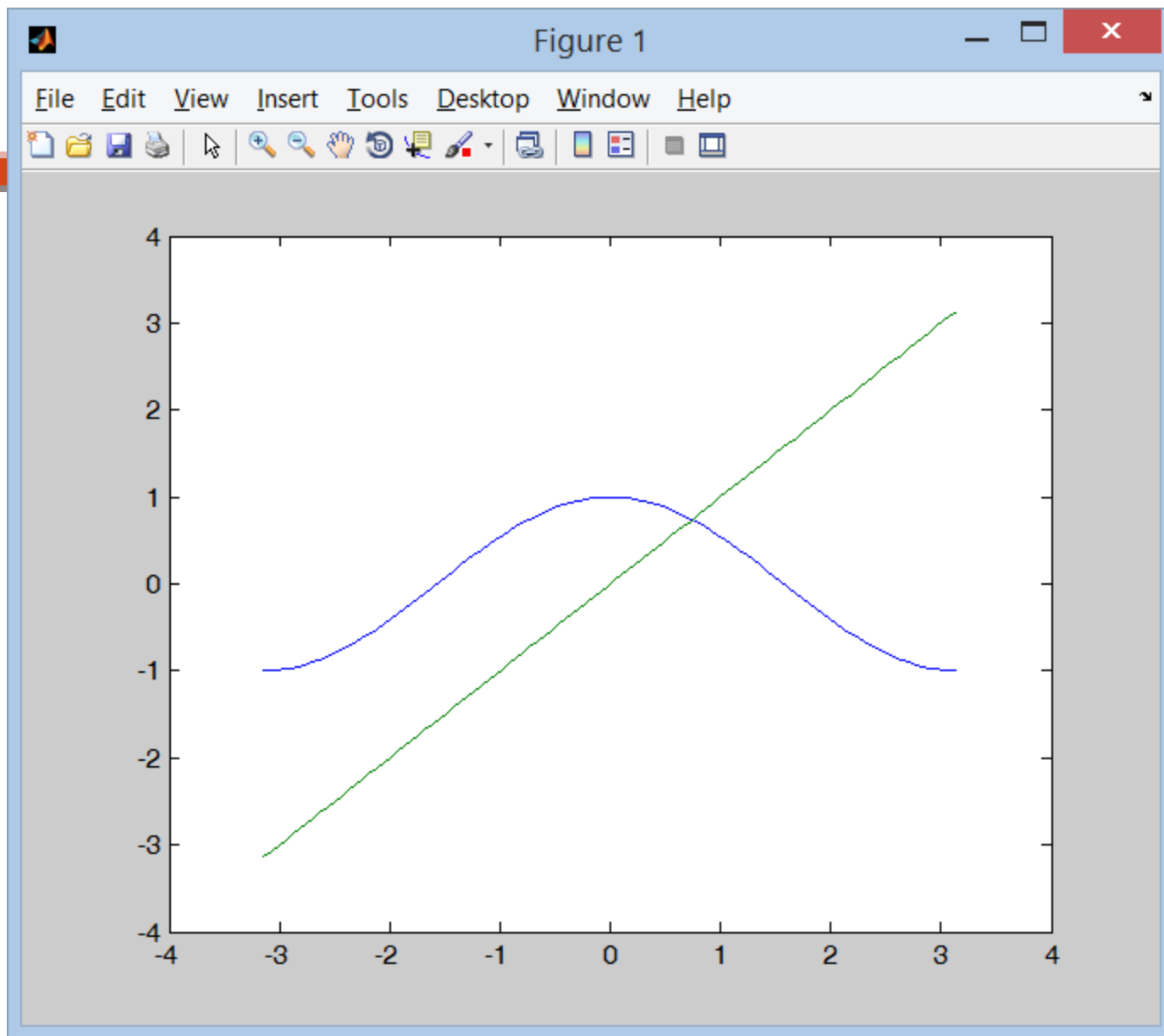
```
plot(xplot,y1plot)
```

```
hold on
```

```
plot(xplot,y2plot)
```

```
hold off
```

```
print -djpeg e1Lab5.jpg
```



Exercise 2

2-(a)

Write an m-file named **cosmx.m** (“COS Minus X”) that defines the function:

$$\mathbf{f(x) = \cos x - x}$$

cosmx.m

```
function y=cosmx(x)
% y = cosmx(x) computes the difference  $y=\cos(x)-x$ 
% your name and the date
y=cos(x)-x;
y=?????

end
```

2-(b)

Use your **cosmx** function to compute **cosmx(1)**.

Result = -0.4597.

```
>> cosmx(1)
```

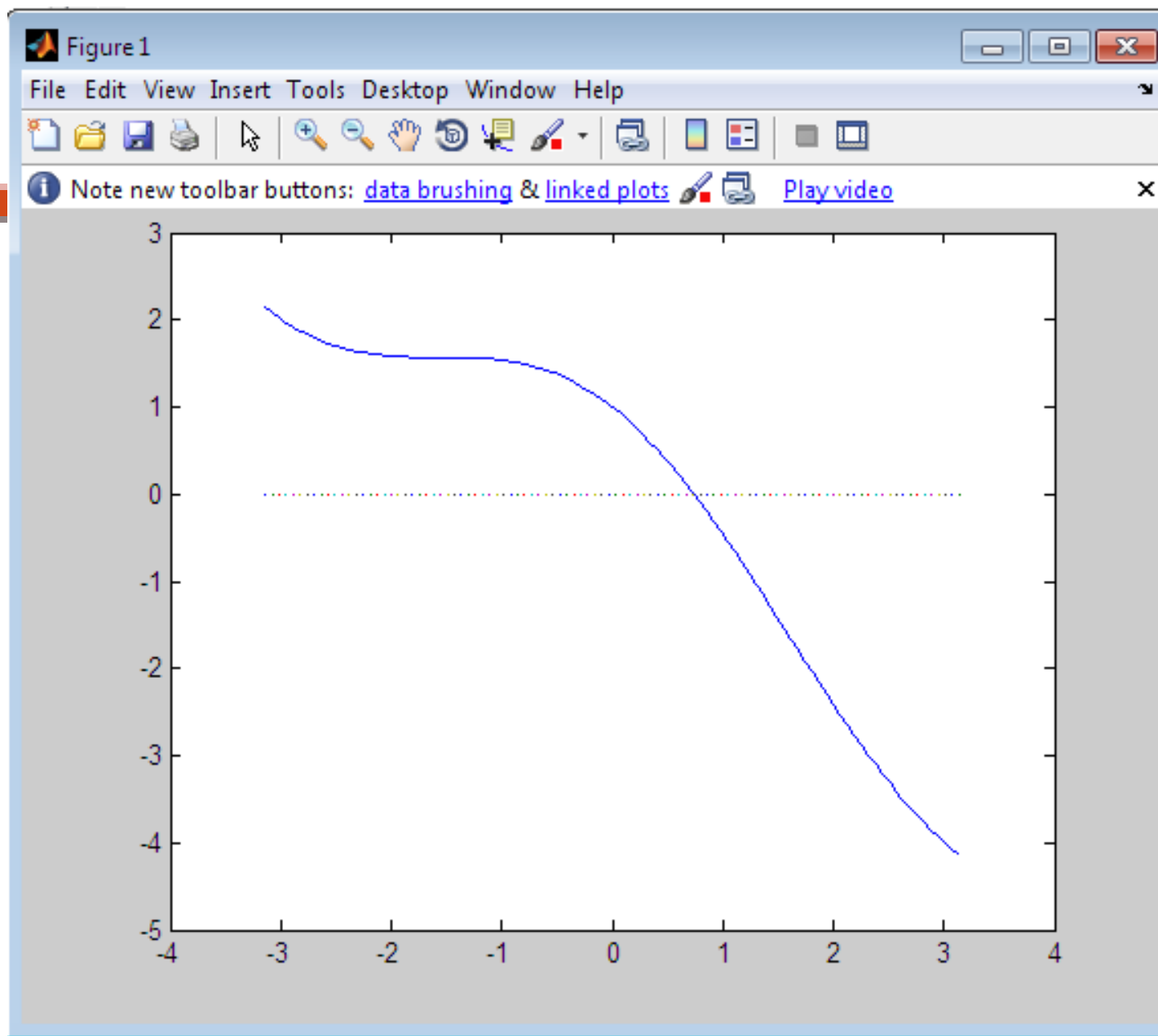

2-(c)

Use your **cosmx** function in the same plot sequence as above to plot **cosmx** and the **horizontal axis** on the interval $-\pi < x < \pi$.

- **Hint:** The horizontal axis is the line $y = 0$.

e2Lab5.m

```
x=linspace(-pi,pi);  
y1plot=cosmx(x);  
y2plot=zeros(length(x));  
plot(x,y1plot);  
hold on  
plot(x,y2plot,'-.');  
hold off  
print -djpeg q2lab5.jpg
```



Exercise 3:

Roots Finding Bisection Method

Bisection Algorithm

```
initialize:  $a = \dots, b = \dots$   
for  $k = 1, 2, \dots$   
   $x_m = a + (b - a)/2$  Another formula for the midpoint  
  if  $\text{sign}(f(x_m)) = \text{sign}(f(x_a))$   
     $a = x_m$   
  else  
     $b = x_m$   
  end  
  if converged, stop  
end
```

- Use both convergence criteria
- Use δ_x and δ_f equal to ε .
(ε is the machine precision)

Convergence Criteria on x

Absolute tolerance: $|x_k - x_{k-1}| < \delta_x$

Relative tolerance: $\left| \frac{x_k - x_{k-1}}{b - a} \right| < \hat{\delta}_x$

x_k = current guess at the root

x_{k-1} = previous guess at the root

$b - a$ = initial bracketing interval

Convergence Criteria on $f(x)$

Absolute tolerance: $|f(x_k)| < \delta_f$

Relative tolerance:

$$|f(x_k)| < \hat{\delta}_f \max\{|f(a_0)|, |f(b_0)|\}$$

where a_0 and b_0 are the original brackets

Stopping Tests (Convergence tests)

After each step for root finding routines, check to see whether the current result is good enough. Common tests include:

- **Residual size** $|f(x)| < \epsilon$
- **Increment size** $|x_{\text{new}} - x_{\text{old}}| < \epsilon$
- **Number of iterations:** $\text{itCount} > \text{ITMAX}$

```

function xm=bisection(f,a,b,n)
%function bisection
%input: f is function ,
%input a,b are the original bracket, n number of iterations
%output: xm the value of root

%set the default number of iterations
if nargin<4
    n=15;
end

%Solve equation using a,b
fa=feval(f,a);
fb=feval(f,b);
%check sign (It should be different)
if sign(fa)==sign(fb)
    error('Root not bracketed by [%f , %f]',a,b);
end

```

```

fprintf('Iterations details:');
k=1;
while k<=n
    %calculating the midpoint
    xm=(a+b)/2;
    %solve equation using midpoint
    fm=feval(f,xm);
    fprintf('k: %f    a: %f    xmid: %f    b: %f    f(xmid): %e\n',k,a,xm,b,fm);

    %check if the equation solved to stop
    if fm==0
        r=xm;
        fprintf('The root is: %f',r);
        break;
    end

    if sign(fm)==sign(fa)    %root lies in interval [xm,b], so replace a
        a=xm; fa=fm;
    else                    %root lies in interval [a,xm], so replace b
        b=xm; fb=fm;
    end

    k=k+1;
end %end of loop

```



```
%Print a warning message if the number of iterations reach the maximum
if k>n
    warning ('root not found after %d iterations',n);
end
end
```

3-(a)

What does the **sign(x)** function do?

What if x is 0?

```
>> help sign
```

What does the **feval(f,x)** function do?

```
>> help feval
```

3-(b)

Type `help bisection` at the Matlab command line.

```
>> help bisection
```

Important Notes

When we use *bisection*, we must include the function name in quotes:

```
xm = bisection ( 'cosmx', 0, 3, 55 )
```

Warning: If you forget the quotes around *cosmx* and use the command

```
xm = bisection ( cosmx, 0, 3, 55 )
```

you will get an error message

3-(c)

Try the command

xm = bisection ('cosxm', 0, 3, 55)

```
>> xm = bisection ('cosxm', 0, 3, 55)
```

3-(d)

Change the value of the variable n to 20, and try the command:

```
xm = bisection ('cosxm',0, 3, 20)
```

What is the output? What does the **warning** function do?

```
>> xm = bisection ('cosxm',0, 3, 20)
```

3-(e)

Try the command:

```
xm = bisection ('cosxm',1, 3, 55)
```

What is the output? What does the **error** function do?

```
>> xm = bisection ('cosxm',1, 3, 55)
```

3-(f)

What are the convergence (stopping) tests used in the function *bisection*.

Homework

- The following table contains formulas and search intervals for four functions. Write four simple function files, `f1.m`, `f2.m`, `f3.m` and `f4.m`, each one similar to `cosmx.m` but with formulas from the table. Then use `bisect` to find a root in the given interval, and fill in the table. Include the table in your report file.

Table

Name	Formula	Interval	approxRoot (r)
f1	x^2-9	[0,5]	
f2	x^5-x-1	[1,2]	
f3	$3*x-\exp(-x)$	[-1,100]	
f4	$2*\cos(3*x)-\exp(x)$	[0,6]	